

HOW DOES MANDATED VOLUNTEERING AFFECT TOTAL VOLUNTEERING?

An Undergraduate Research Scholars Thesis

by

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ABSTRACT

How Does Mandated Volunteering Affect Total Volunteering?

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Organizations, non-profit or otherwise, often rely on the work of volunteers to accomplish their goals. And although it is often the case that sufficient volunteers can be procured to complete a given task, it remains an open question as to the most efficient means of attracting volunteers to devote their time and effort to a cause. This study builds upon previous research investigating the demographics of subsequent and long-term volunteerism by expanding on how requests to volunteer are framed: either as events for which one can choose to volunteer however much of their time as desired, or as “mandated” volunteering, where participants are expected to devote a certain minimum amount of time, but are then free either to leave or to continue volunteering as they please. The question is then simply which method achieves the highest amount of total hours worked for an individual. Our study is based on an experimental segment to be conducted in the lab, which derives meaningful conclusions on the most efficacious methods an organization can employ in order to increase their events’ effectiveness. We find that mandated volunteering has negligible effects upon one’s decision making when volunteering, assuming that said mandate is at or near a supposed equilibrium level of volunteering.

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CHAPTER I

INTRODUCTION

When an event needs to be staffed, or a cause addressed, it is not uncommon to “mandate” members of organizations, clubs, or businesses to provide volunteers to contribute to those efforts. However, the question of whether compelling people to volunteer is equally effective, less effective, or more effective than simply requesting potential participants to give of their time freely and of their own motivation has yet to be firmly answered. Although the concept of mandated volunteering is increasingly well researched in the field of experimental economics, the efforts of researchers have been primarily directed towards the long-term effects of mandated volunteering on an individual’s propensity to volunteer, rather than its immediate effects on, for example, net time contributed. In this study, we seek to answer the latter question, seeing it as fundamental to the outgrowth of questions that hinge on mandated volunteering’s actual use. In essence, we wish to establish whether or not mandated volunteering is an efficient use of volunteers’ time and willingness to volunteer, in an experimental setting, or whether mandated volunteering “crowds out” a volunteer’s level of altruism. We anticipate that a mandate will indeed crowd out altruism and decrease average individual volunteering time.

In this paper, we employ a novel experimental paradigm, the internet-based charity game “Freerice.” Freerice is a non-profit website owned by and in support of the United Nations World Food Programme (UNWFP). Its stated goals are to “provide education to everyone for free” and to “help end world hunger by providing rice to hungry people for free” (“About Freerice.com,” n.d.). The website takes the form of an interactive quiz terminal, where participants answer an

unlimited series of questions in a subject matter of their choice. For each question correctly answered, ten grains of rice are donated to people in need. We chose Freerice as the basis for our experiment because it closely approximates the actual environments in which volunteering takes place. Namely, it has a clear charitable goal, an element of entertainment, and the opportunity to choose a subject matter of an individual's personal preference. In addition, the Freerice platform allows for easy replicability and a greater preponderance of control than would be possible in the field.

We put our question to the test using a between-subjects analysis of a control group, where no mandate existed, and two groups with different time mandates. Results indicate a difference between the control group and one of the two mandates, but a less clear distinction between the control group and the second. In light of this finding, the timing of the mandate becomes particularly important. If the mandate is quite below the "equilibrium" level of volunteering, then relying on an average individual's agency may be a better way to optimize the amount of work completed.

Literature Review

Previous research into the effect of mandated volunteering on total volunteering is scarce, and it is only recently that the topic has received much attention in the experimental field. Where such prior research exists, we incorporate it into this paper, but for the most part, we are required to derive our background information indirectly. Despite the relative lack of research pertaining to immediate effects of mandated volunteering, a great deal of research has been conducted into the

long-term effects of the practice. We begin with a recount of this literature, and we move into a discussion of mandated volunteering's immediate effects afterwards.

In 1966, Jack W. Brehm developed his “reactance theory.” This theory posited that when an individual is compelled to perform a task or engage in an activity, they will perceive a corresponding loss of personal freedom and independence that will lead to a negative reaction to whatever mandate they have been given. They will then rebel against the activity or task, sometimes even taking the opposite reaction to it (Brehm, 1966). Likewise, in 1973 researchers looked into the role of extrinsic motivators in subsequent interest in a given activity. This early research indicated that extrinsic motivators, when made known to the subject before they engage in an activity, decrease the subject's subsequent intrinsic motivation to engage in the activity in comparison to subjects who received an award afterwards or who were given none at all (Lepper, Greene, & Nisbett, 1973). Similarly, our experiment compared groups who were told ahead of time that they could leave after a given period of time had elapsed (the extrinsic motivation/external mandate) and a baseline group whose subjects were not given a specified, “satisfactory” time.

Alongside extrinsic motivators, the mere presence of external forces also poses interesting questions. Past research has found that when an individual believes that their actions are subject to external forces, this undermines their own sense of self-determination and, critically, lessens the association of the individual's activities with their own intrinsic motivation (Deci & Ryan, 1985). Similarly, Kruglanski demonstrated that the presence of an external driver may give an individual the perception that they are being compelled to do something they might not otherwise

do. This decreases their interest in the activity and leads them to experience less enthusiasm for the activity than they otherwise might have (Kruglanski, 1978).

As mandated volunteering became more common in the 1990s, particularly in the public education system, questions began to emerge both to its efficacy in inspiring students to view civic service positively as well as its legality as a form of “coerced labor.” In his review of the practice and its psychological effects in *Law & Psychology Review*, Sobus suggests that alternative means be found to develop civic virtues in students, saying “that using external constraints to compel behavior can have the boomerang effect of actually decreasing the desired behavior when the constraint is removed” (Sobus, 1995). He goes on to assert that “a coercive policy should be expected to undermine positive attributions, stifle feelings of self-determination, and ultimately make self-generated acts of community service more scarce” (Sobus, 1995). In our experiment, we sought to determine if similar effects could be observed “in the moment” by comparison of externally constrained subject behavior and subjects who were only internally motivated.

Although the theoretical backdrop for mandated volunteering was already well established, field and laboratory studies of the practice have emerged only within the last two decades. Helms, Angner, Scott et al. developed an experimental approach to volunteering in their paper published in 2011. Subjects played a computer-based game in which they pressed a button on the screen. For every minute they did so, \$0.20 was transferred to a charity. Their results indicated that a mandate does indeed seem to crowd out volunteerism, even to the point of reducing total time volunteered (S. Helms, Angner, Scott, & Culver, 2011). However, we believe that their

experimental design may have included too much phrasing that “primed” subjects to leave earlier than they otherwise might have. We believe that this may have had the effect of creating an environment in which subjects felt that it was “socially acceptable” to leave once the mandate had expired, or at any time at all, in the case of the non-mandated baseline. In our experimental design, we seek to remedy these and other issues, such as the experimental task itself. The experimenters acknowledge in their paper that the volunteering game they designed is boring and repetitive. This was intentional in their case, as they sought to eliminate the extraneous effects of “inherent enjoyment.” In our design, we employ a vehicle which is more immersive because we find that this may be a more representative indicator of behavior in the environments in which volunteering actually takes place.

In a 2013 study however, Helms’ own investigation of restricted-access data from Maryland public schools found that the state high schools’ practice of requiring a certain number of volunteer hours in order to graduate increased the level of volunteering amongst eighth graders, but decreased it amongst twelfth graders. Here, she underscored the possibility that mandated volunteering may change only the timing of volunteerism, and not the actual total amount volunteered (S. E. Helms, 2013).

A 2007 field study from Canada took advantage of a reform to the high school system in Ontario in which the length of education was abbreviated by one year for the incoming class, but who were also compelled, unlike their older peers, to perform a certain amount of volunteering hours. Here, the researchers found that the volunteering mandate had the effect of increasing later

volunteering in comparison with the other cohort (Pancer, Brown, Henderson, & Ellis-Hale, 2007).

Mandated volunteering finds its most tangential corporate application in the legal field, where *pro bono publico* (generally referred to simply as “pro bono”) hours are a respected and growing part of a lawyer’s practice. For most of US legal history, pro bono hours have been seen as a voluntary contribution of a lawyer’s time and resources to individuals who would otherwise be unable to pay for their legal services. However, there has been a recent trend towards making pro bono hours a mandatory part of a lawyer’s profession. For example, in the American Bar Association’s Model Rules of Professional Conduct, it states that “every lawyer has a professional responsibility to provide legal services to those unable to pay. A lawyer should aspire to render at least (50) hours of pro bono publico legal services per year” (American Bar Association, 2013). In 2012, the New York State Court of Appeals issued a new rule requiring that applicants to the New York Bar complete 50 hours of pro bono service as a prerequisite for admittance (New York State Unified Court System, n.d.). Although no state has yet made pro bono hours mandatory on an annual basis, some jurisdictions, such as Florida and Maryland, have implemented a mandatory reporting system (Boyle, 2007). In those states, Leslie Boyle, of Georgetown University Law Center, reiterates that those states have seen substantial increases in pro bono volunteering hours since the introduction of the mandatory reporting systems, but it is not clear from the paper how much of those increases are attributable to the new reporting system and not to wider changes in the pro bono field (Boyle, 2007). Nevertheless, Boyle argues that a mandatory pro bono service requirement would increase both the numbers of lawyers who volunteer and the number of hours volunteered.

The comparison of mandated volunteerism to pro bono hours is not without some drawbacks. The two are not one hundred percent compatible with one another. Whereas a case could be made that volunteerism (including pro bono volunteering) in general is a social duty—we will not attempt to prove or disprove this, merely note its existence—most volunteering does not entail thorny questions of monopoly power in the same way that the legal field does. It is a notable characteristic of the legal field that its barriers to entry are far higher than, for example, working for Habitat for Humanity or fundraising for cancer research. Only lawyers may practice the law, and so only lawyers can volunteer legal services. Lisa Tudzin notes this distinction and argues that a lawyer's monopoly power obligates them to provide pro bono services (Tudzin, 1987). Although generally perceived to be a kind of volunteering, the differences between pro bono service and regular volunteering may be significant enough to warrant a separate study due to the different social pressures felt by an attorney.

In general, there appears to be a lasting uncertainty in the field of volunteerism that has defied attempts to develop a consistent theoretical model that describes its changes in the context of a mandate. In this experiment, we apply our own experimental model in an attempt to offer a new perspective and refinement on the issue that these previous papers have addressed.

CHAPTER II

EXPERIMENTAL DESIGN

Our experiment is a between-subjects design, in which a group of subjects completed a baseline experiment, and then two other groups of subjects subsequently performed the same experiment with only one factor (the time mandate) altered from the original baseline.

In the baseline experiment, registered subjects were allowed to begin at any time during a three-and-a-half-hour window, from 1:30 PM to 5 PM. Subjects signed in at the front window of the ERL, and the experimenter took note of their entry time. Subjects then read an information sheet, and were subsequently taken into the lab, and seated at a computer pod. The experimenter gave the subject an entry survey, where they were asked to fill out some demographic information. We took care to avoid any questions that might have primed the subjects to respond in any way towards charity, gameplay, or volunteerism in general (as seen on page 30, Appendix B). The exit survey contained questions relating to subjects' volunteering history and beliefs and was administered after the subject elected to end their session.

Upon completing the survey, the subject put on over-head ear phones, and was played a short video with audio narration, which was displayed on the Windows Media Player application on their pod's computer screen (refer to pages 35-55, Figures 12-15). This video explained that the subject would be performing an act of volunteering by participating in the UNWFP's program, Freerice. The video goes on to explain the mission of Freerice and instructs the subject how to maneuver through the website's interface so that they are comfortable with the site's layout and

functioning, and to ensure that their activities would submit valid and useable data. The video mentioned that subjects will receive a fixed payment of \$15 and explained that subjects are allowed to volunteer as long as they please and that their reimbursement is in no way contingent upon the length of their volunteering. In closing, the video instructed subjects on how they may proceed after completing their volunteering session. The text of each of the videos is included in Appendix B. It should be noted that we made one small change to the control group video after we observed a situation in which one subject was confused on how to end the experiment. Although we ran a full session using this original baseline video, we do not use the data for interpretation in our analysis and conclusion sections. This change is well documented in Appendix B, and can be seen most clearly in the differences between the “Lab Regulations” slides on pages 38 and 42-43.

After watching the video, subjects then exit the video application and find the Freerice webpage already opened and ready for them to begin volunteering. We gave subjects anonymous, single-use usernames, which the experimenters had already started up and prepared in advance. Subjects were requested not to use any preexisting Freerice account that they might have, which, thankfully, none did. Each subject was then given the opportunity to answer questions from a myriad of topics including humanities, English, mathematics, and many more. As mentioned, the baseline experiment allowed subjects to donate as many or as few grains of rice as they wished. They were also similarly permitted to stay as long as they pleased.

When the subject determined that they were finished, the subject raised their hand, and the experimenter came to the subject to offer assistance. If they requested to leave, the experimenter

noted the time and the subject's username on an exit survey and gave it to them to fill out, asking the subject to return it to the administrative experimenter when finished. This exit survey asked questions about the subject's prior experience pertaining to volunteering (page 32, Appendix B). Once completed, the subject left the lab and returned to the payment window, where they signed out and receive a fixed payment of \$15.

The following treatments mimicked the baseline, but slightly altered one factor, the time mandate. After the original baseline treatment ended, we calculated the subjects' average time spent in the lab. From these statistics, we determined the "mandate" for subsequent treatments. Two treatments followed the baseline, which mandated subjects to volunteer for a fixed amount of time (roughly 20% and 40% of the baseline average time spent). The choice of 20% and 40% mandates were partially arbitrary. On the one hand, we wanted lower mandates so that we could more easily perceive differences between the baseline and the treatment sessions, but then again, we also realized that there would be little meaningful difference between 40% and 39% mandates. For this reason, we did not alter the mandated times after we threw out the first baseline session on which these numbers were based, recognizing that this would make little difference to our final results. These time mandate caveats were expressed in each treatment to the subject by the video that they were asked to watch before volunteering (see Appendix B, pages 44-55). In addition, these treatment sessions were run at the same time of day as the baseline. The subjects in this experiment were at no point *forced* to volunteer. The presentation of the experiment expressed the given time as a "request" rather than a "mandate." It was not made clear to the subjects whether or not there were consequences within the game for leaving early, even though there were, in fact, none. We constructed this ambiguity intentionally in order

to replicate real life situations of mandated volunteering, in which organizations often “request” that their members volunteer a certain amount of time or until a certain task is accomplished, but where, generally due to peer or other social pressures, this request is understood as a “mandate.” Subjects were allowed to leave the session whenever they pleased, and were neither reminded of the mandate when they requested to leave nor subjected to a decreased payment upon checking out.

Data was recorded using the Freerice usernames, which were linked only to the subjects’ anonymized entry and exit surveys. Following the close of all experimental sessions, we began analysis of the collected data.

CHAPTER III

RESULTS

The results of the experiment suggest a statistically significant difference at the 20% mandate in our t-test and in all our OLS regressions. Results for the 40% mandate did not attain statistical significance at any level. At the 20% mandate level, participants volunteered on average 7.3 minutes less than in the baseline (lowest average). At the 40% mandate level, volunteers participated on average 2 minutes longer. It should be pointed out however, that the 40% mandate time (24 minutes) was based on our original baseline average of 60 minutes of volunteering. Had the mandate instead been based off of the second, lower average time, the 40% mandate would have been 7 minutes shorter (17 minutes). There are strong grounds to believe that the results of the 40% mandate may have been different had the participants been subject to a mandate based on the time their subject group would eventually have been compared to, rather than a different number entirely.

We will now proceed into further detail regarding the specifics of these results, broken down into individual treatments, demographics within treatments, and some general remarks about the character of the data distributions.

When the 20% mandate was tested against the observed mean volunteering time of the 2nd baseline (see Table 2, Appendix A), the 20% mandate participants volunteered on average 7.33 minutes less, attaining statistical significance at a 5% level. However, when the 40% mandate

was tested against the 2nd baseline (Table 3, Appendix A), the latter's subjects volunteered on average 2.3 minutes more, though this difference was not statistically significant at any level.

In addition to time volunteered, we also collected a series of demographic variables to further break down our results. However, we found that, across ten different regressions, only age had a statistically significant effect, with an increase in age having a positive relationship with minutes volunteered. However, we note that this observation is unlikely to be reliable, despite its statistical significance. Because of our laboratory's physical location on a college campus, there was very little variation in our age range in the first place, so we suspect that this result may be somewhat misleading. Nevertheless, the demographic indicators were broadly in keeping with our belief that the presence of a mandate would prove to be the dominant effect on volunteering times. In Table 4 of Appendix A, the precise results of this regression can be viewed in greater detail. We define each of our variables as follows. "Twenty" refers to participants in the 20% mandate treatment group. "Forty" refers to participants in the 40% mandate treatment group. "Age" is the participant's age in years. "Female" indicates whether the participant identified herself as a female. "Past Volunteer" indicated whether or not the participant had past experience volunteering. "Liberal" indicated if the participant identified his or her political ideology as left-wing or liberal. "White" indicated if the participant identified as racially or ethnically white and non-Hispanic. "Part Time" indicated if the student was attending Texas A&M University on a part time basis. "Class Year" represented how many years the student had attended Texas A&M University on a sliding scale of 1 to 6. "Religious" represented those participants who attended religious services at least once a month. One final note: we established a variety of cutoffs for "religiousness" in several separate regressions that we ran. As Helms, Angner, Scott et al. note in

their paper, religiousness was found to be an important factor in volunteering, and as such, we were eager to see if we could find any effects of our own (S. Helms et al., 2011). Surprisingly, no matter how we defined the term, we could find no evidence of any statistically significant impact in our own study.

Although demographics shed no light on the behavior of participants, a look at the distributions may hint at a possible, though as yet unexplored, explanation. In Figures 5 through 9, and particularly in Figure 9, cumulative distribution functions can be seen for each of the different treatments and each of the baselines. It is immediately noticeable that the slopes of the baselines and the slopes of the experimental groups appear rather different. Whereas the baselines seem to indicate a steady rate of disengagement amongst participants, in each of the experimental groups, the rate of disengagement increases sharply about ten minutes after the expiry of the mandates (12 and 24 minutes). After this point, their slopes continue to be much steeper than those of the baselines until the last participants begin to trickle out. We are currently working on statistical means to distinguish between the different distributions in order to put forth a more credible assessment of their relative worth without defaulting to less reliable visual observations.

CHAPTER IV

CONCLUSION

The key takeaway from our data is that the mandate does not censor behavior. This is contrary to our initial expectations that the presence of a mandate would indicate a strong kind of implicit “permission,” or even encouragement, for the participants to leave the study. At best, we instead observe a possible, and much weaker, kind of influence, where participants increase their rate of disengagement after a mandate’s expiry, but on average do not leave much earlier except when the mandate is set to expire long before the baseline average volunteering time (in the 20% mandate). Nevertheless, altered disengagements would be noteworthy if provable. These changes would indicate that the presence of a mandate *does* have a signaling effect on participants, though not one so strong as to cause an entire cohort to vacate the laboratory within a few minutes.

Although we remain surprised that the significance of the differences between the baselines and the experimental treatments was so weak, we can guess at possible causes related to the design of our experiment rather than to “typical” volunteering behavior.

First, the difference between average volunteering time and the two mandated times was not all that great. Although to a college student, the difference between 12 minutes of volunteering and an hour of volunteering might seem immense, statistically, these numbers are problematic. Due to our small sample sizes, the possible spans of time may not have been large enough to highlight real underlying differences between mandated and un-mandated volunteering behavior. When

one considers that relatively few volunteering projects last less than *several* hours, it becomes clear that one single hour may not be all that representative of actual volunteering practice. Due to laboratory constraints however—keeping busy college students in a laboratory for half the day is quite infeasible—it is unlikely that a more accurate setting for volunteering could be achieved anywhere but the field, which poses its own unique organizational challenges. Nevertheless, field studies could provide useful distinctions to our results, especially since the higher disengagement rates after a mandate’s expiry hint at greater average volunteering time differences over longer periods.

Second, volunteers may have been put off from leaving earlier out of consideration for our laboratory assistants. At the mandate’s expiring, the laboratory generally had more than a dozen students in the room, all of whom were served by just two lab assistants. If two students raised their hands to indicate that they were ready to depart, other participants may have waited to indicate their own readiness to leave until a laboratory assistant was obviously available to come and help them. Although this would seem like a constant effect across all sessions (participants in the baseline groups were subject to the same requirements), the difference in the rate of disengagement between the baselines and the experimental groups would likely cause the delay to be felt more acutely in the experimental groups. If we are correct in this hypothesis, we can surmise that the rate of disengagement in the experimental groups would have been even greater, which could have caused average volunteering times to attain statistical significance. We could possibly have remedied this by having more available laboratory assistants, but this proved to be organizationally difficult. Additionally, having too many personnel in the lab at any given time may have confused the participants or even signaled to them that we *expected* many people to

leave at once, potentially creating a strong experimenter bias. Once again, the best solution to this problem would be a field study.

Third, when we were first designing this experiment, we lacked the means to begin timing a participant's volunteering time right when they began interacting with Freerice. Instead, we were forced to use a proxy: beginning their time when they signed in and entered the lab. This is not ideal, although we surmised that the effect would not be too destructive to the validity of our experiments because the effects would be constant across treatments. It took about ten minutes for participants to get set up in their pod in all treatment sessions and in the baseline. We are now working on ways to remedy this situation by showing participants the video and having them fill out the survey ahead of time, rather than after the time has already begun in the lab.

Reconciling our own results with those cited in the literature review is problematic for a variety of reasons. Of course, different population demographics, different timings and settings, or more or less interesting volunteer work could all skew the results of a particular study in one way or another. Rather than focusing on whether or not a mandate leads to decreased aggregate volunteering, we believe it is best to look to the different placements of the mandate. We observe that this placement has a very significant effect on whether or not a mandated participant volunteers longer than a non-mandated participant. Regardless of the results we find in our study, a mandate set at 10% of the average volunteering time will intuitively lead to less total volunteering than one set at 200%! When one does consider our results, it becomes clear that choosing where to set the mandate governs volunteering time more so than any other factor.

Put simply, our results indicate that mandates seem to change the timing of volunteering behavior. Looking back at our literature review, we see that the studies referenced held divergent results. Helms, Angner, and Scott's laboratory experiment indicated that a mandate decreased total volunteering, whereas Pancer, Brown, Henderson, and Ellis-Hale's Canadian field study indicated that a mandate increased total volunteering. Given the aforementioned difficulty of reconciling our own study with any other study in particular, we instead offer a general reconciliation between all studies that involve mandated volunteering, that the timing or placement of the mandate has a determinative influence on average volunteering time due to the strong differences observed between the 20% and 40% mandates.

As mentioned previously, we believe that the field of mandated volunteering would benefit most directly from a new field study. However, rather than hitching a ride on another organization's volunteer drive, we would encourage future researchers to develop their own volunteering program in cooperation with a charity or other organization. Our reasoning for this stems from the need, as yet unmet in the literature, for a field study in which the mandate is determined by the researchers so that differences in disengagement rates can be more narrowly and directly targeted. Rather than focusing on total volunteering time, which we see now relies very clearly on one factor above all others, looking to the rate of disengagement could provide a better picture of how volunteers respond to the presence of a mandate. We understand, of course, the difficulty of organizing such a field study, but we believe that the dividends may well be worth the effort.

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APPENDIX A

TABLES AND GRAPHS

Table 1: Summary Statistics for Treatments

Variable	Obs.	Median	Mean	Std. Dev.	Min.	Max.
1 st Baseline	25	58	59.84	27.95	14	107
2 nd Baseline	36	46	44.31	15.53	14	70
20% Mandate	42	34	36.98	12.76	20	76
40% Mandate	44	45.5	46.61	10.99	24	77

Table 2: Test for 20% Mandate

<i>Two-sample t test with equal variances</i>						
Group	Obs.	Mean	Std. Err.	Std. Dev.	[95% Conf. Interval]	
2 nd Baseline	36	44.31	2.59	15.53	39.05	49.56
20%	42	36.98	1.97	12.76	33.00	40.95
Combined	78	40.36	1.64	14.48	37.09	43.62
Difference		7.33	3.2		.95	13.71
Diff = mean (2 nd Baseline) – mean (20%)				t = 2.2886		
H ₀ : diff = 0				degrees of freedom = 76		
H _a : diff < 0		H _a : diff ≠ 0		H _a : diff > 0		
Pr (T < t) = 0.9876		Pr (T > t) = 0.0249		Pr (T > t) = 0.0124		

Table 3: Test for 40% Mandate

<i>Two-sample t test with equal variances</i>						
Group	Obs.	Mean	Std. Err.	Std. Dev.	[95% Conf. Interval]	
2 nd Baseline	36	44.31	2.59	15.53	39.05	49.56
40%	44	46.61	1.66	10.99	43.27	49.96
Combined	80	45.58	1.47	13.19	42.64	48.51
Difference		-2.31	2.97		-8.22	3.61
Diff = mean (2 nd Baseline) – mean (40%)				t = -0.7768		
H ₀ : diff = 0				degrees of freedom = 78		
H _a : diff < 0		H _a : diff ≠ 0		H _a : diff > 0		
Pr (T < t) = 0.2198		Pr (T > t) = 0.4397		Pr (T > t) = 0.7802		

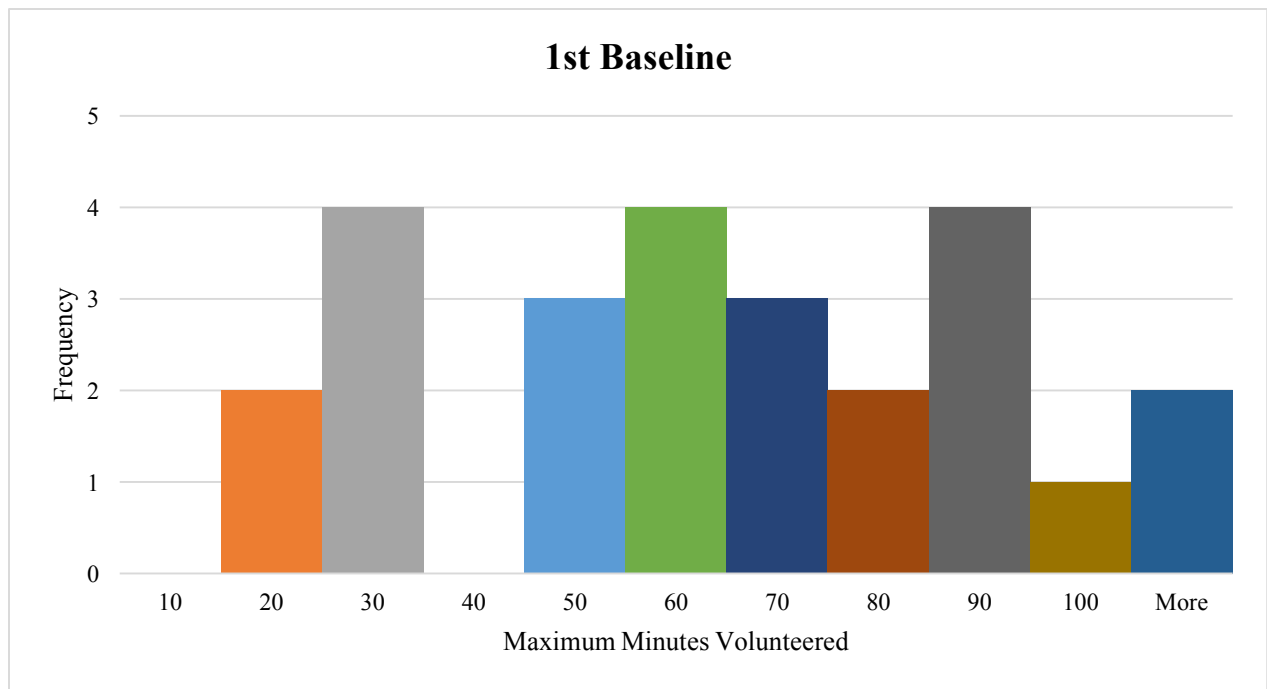


Figure 1: 1st Baseline Histogram

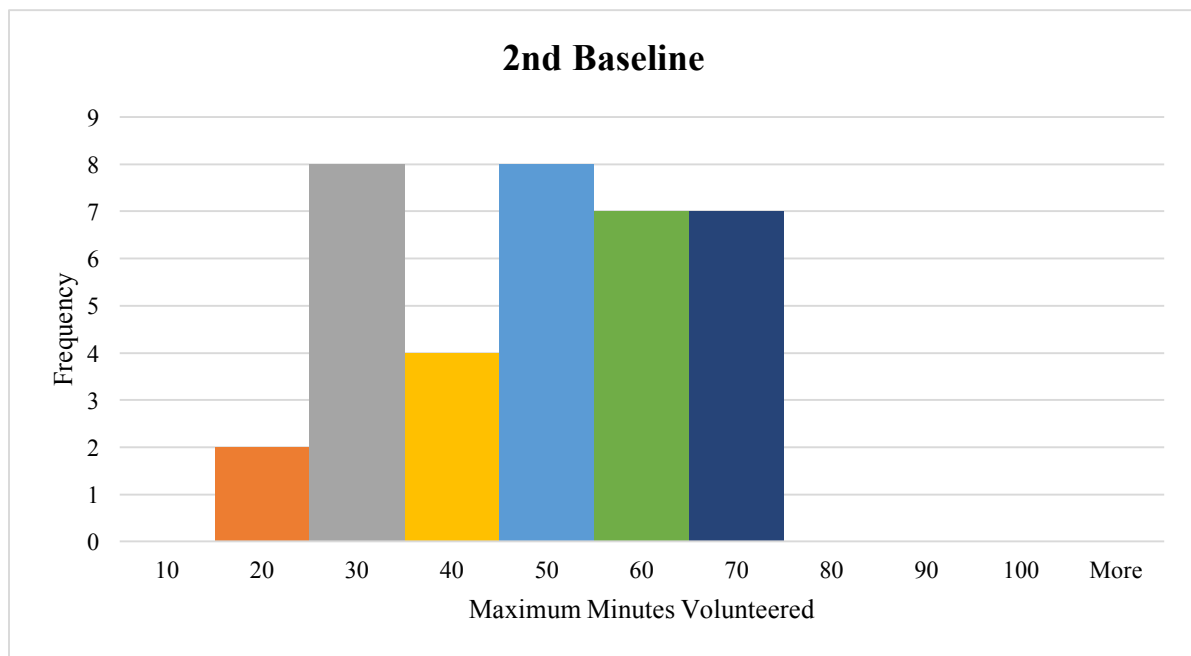


Figure 2: 2nd Baseline Histogram

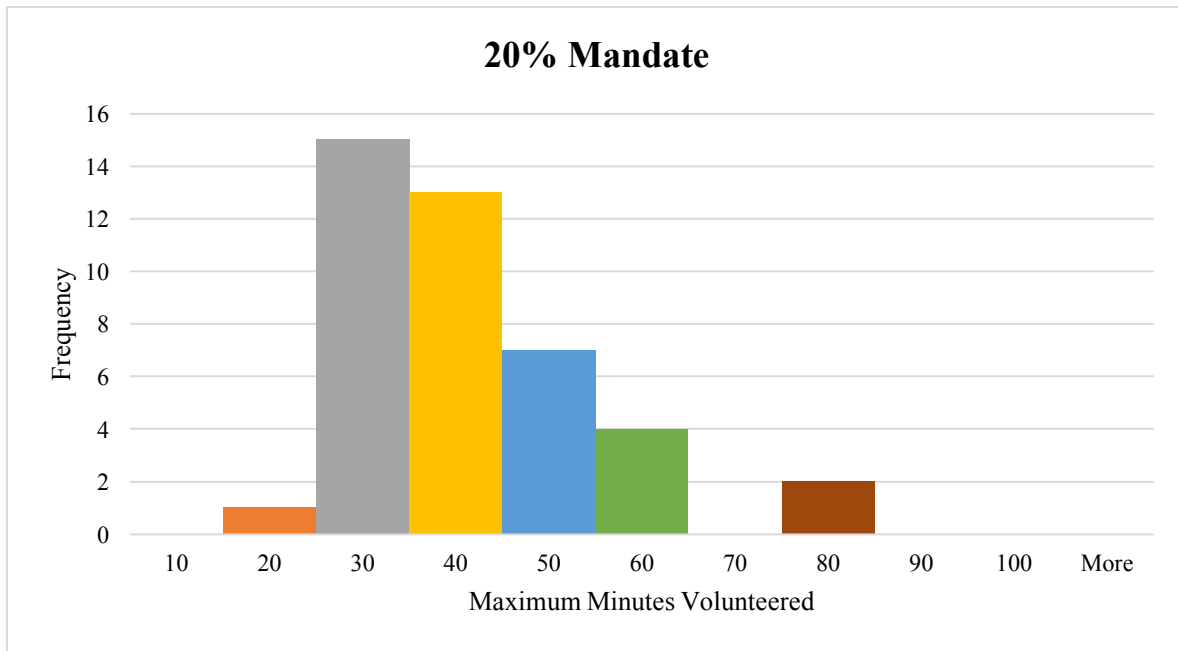


Figure 3: 20% Mandate

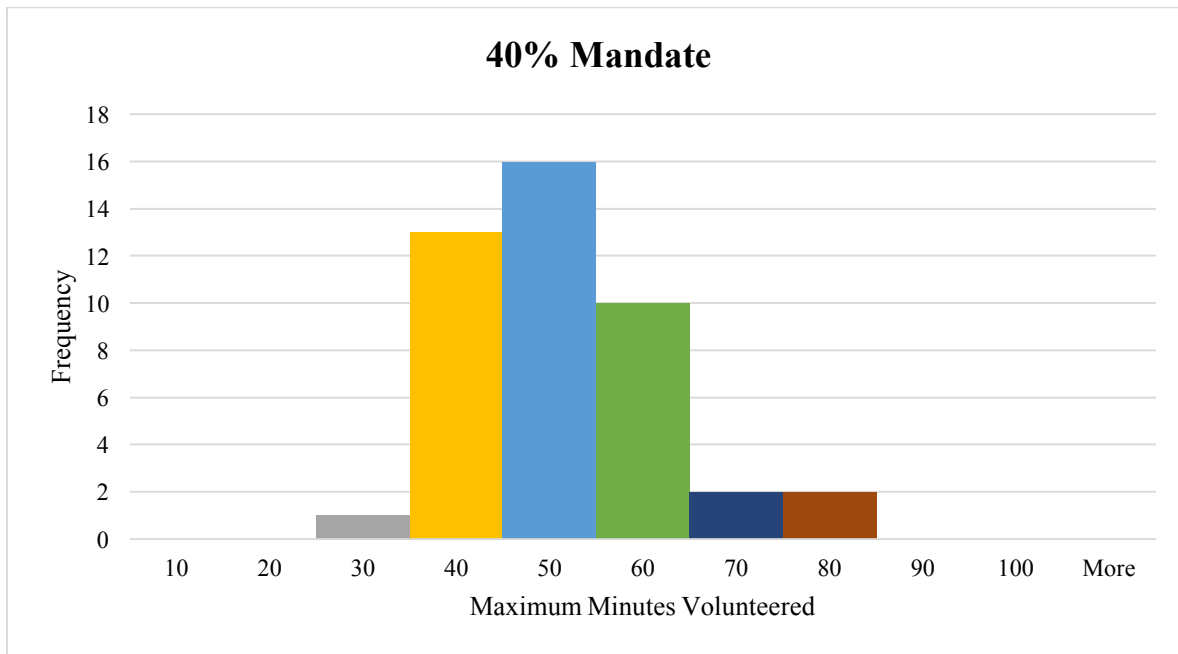


Figure 4: 40% Mandate

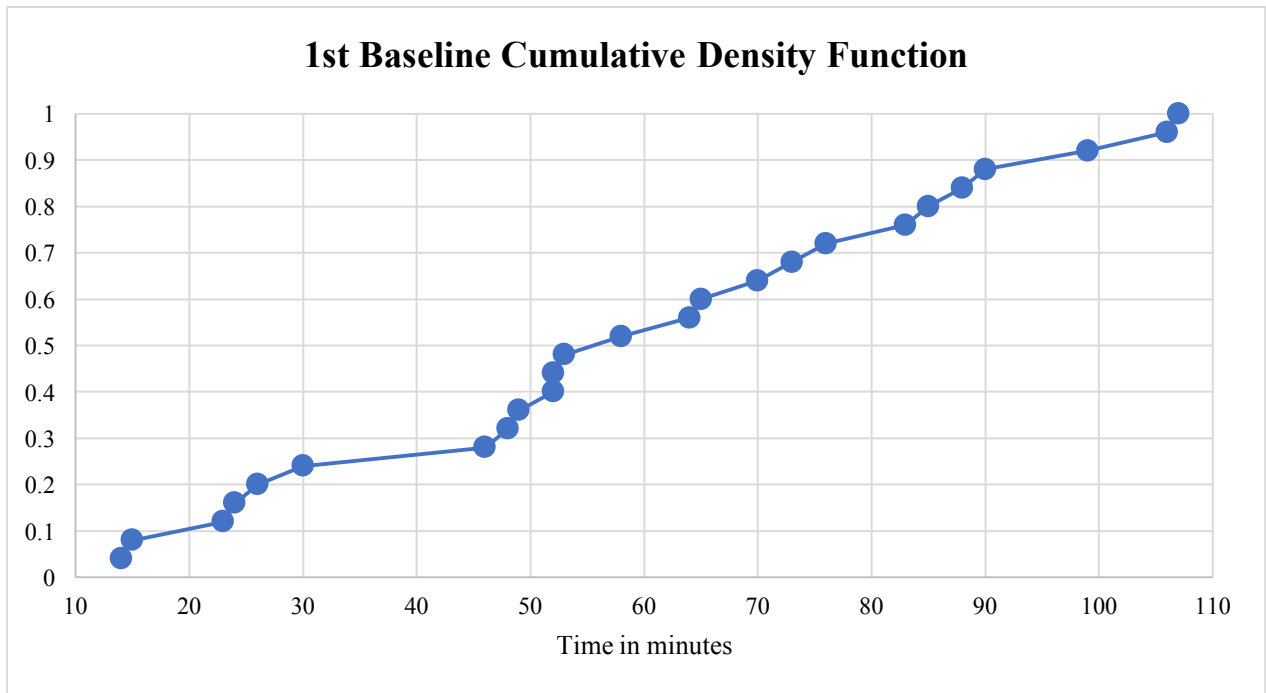


Figure 5: 1st Baseline Cumulative Density Function

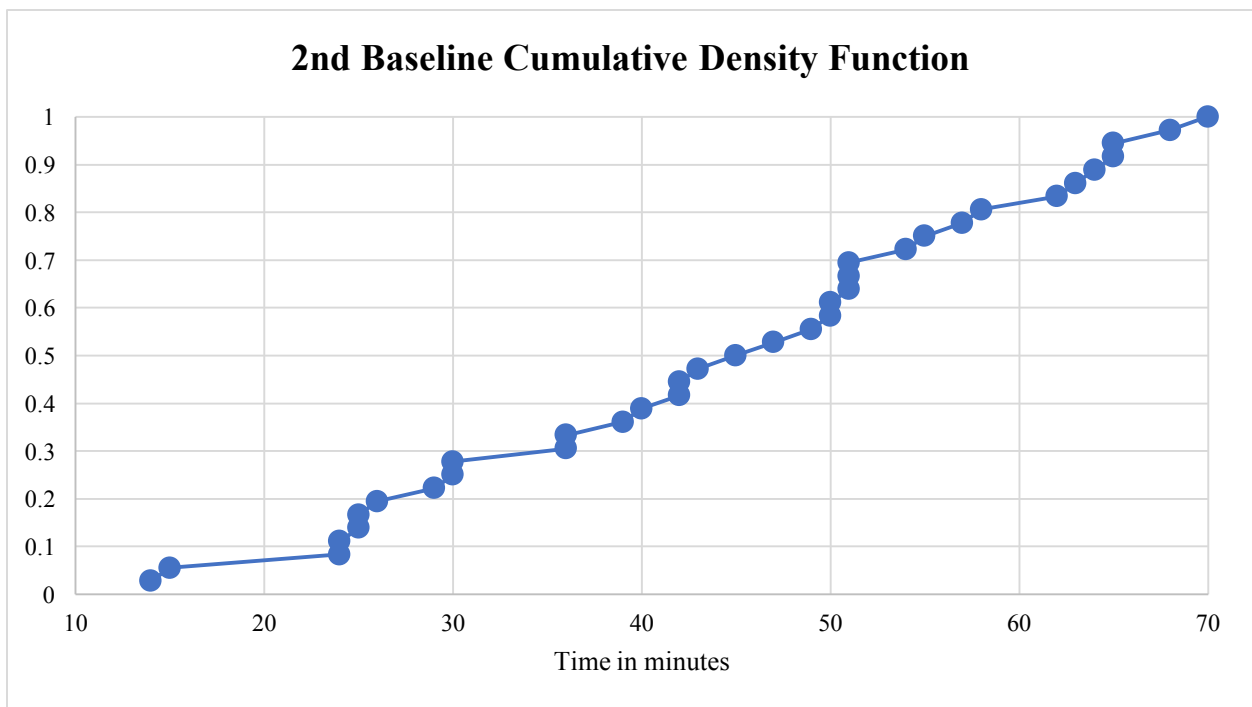


Figure 6: 2nd Baseline Cumulative Density Function

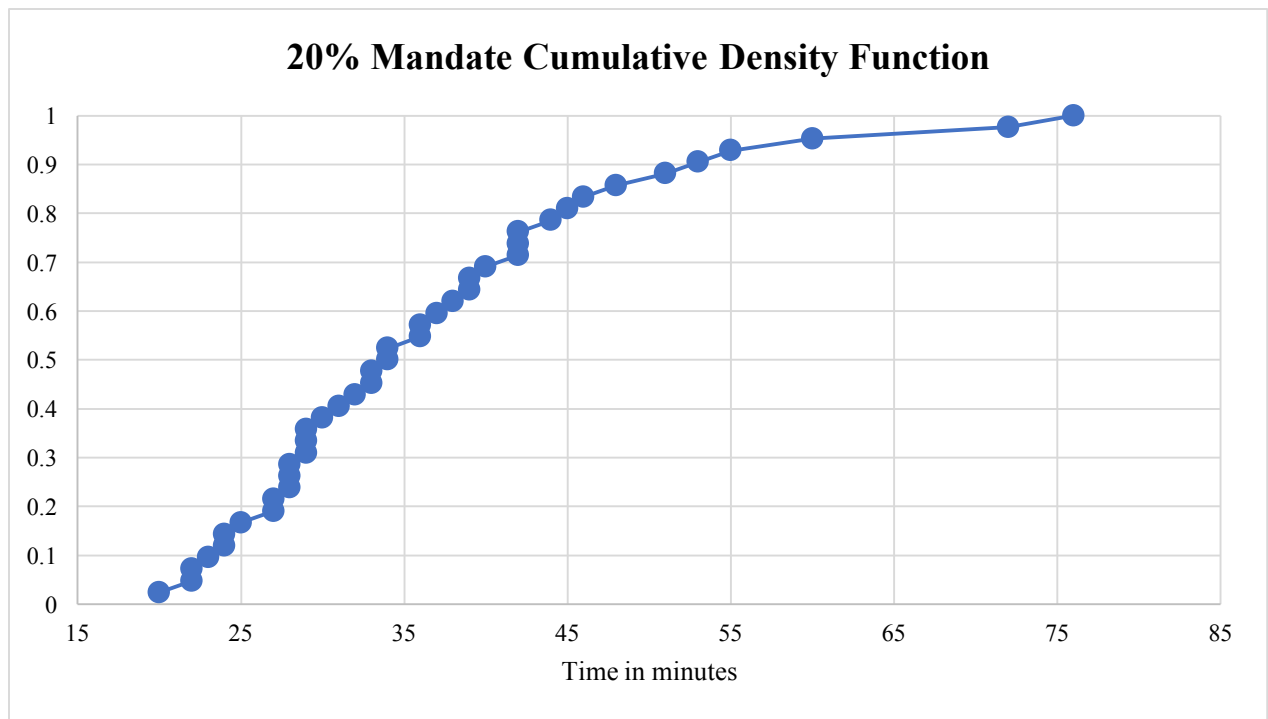


Figure 7: 20% Mandate Cumulative Density Function

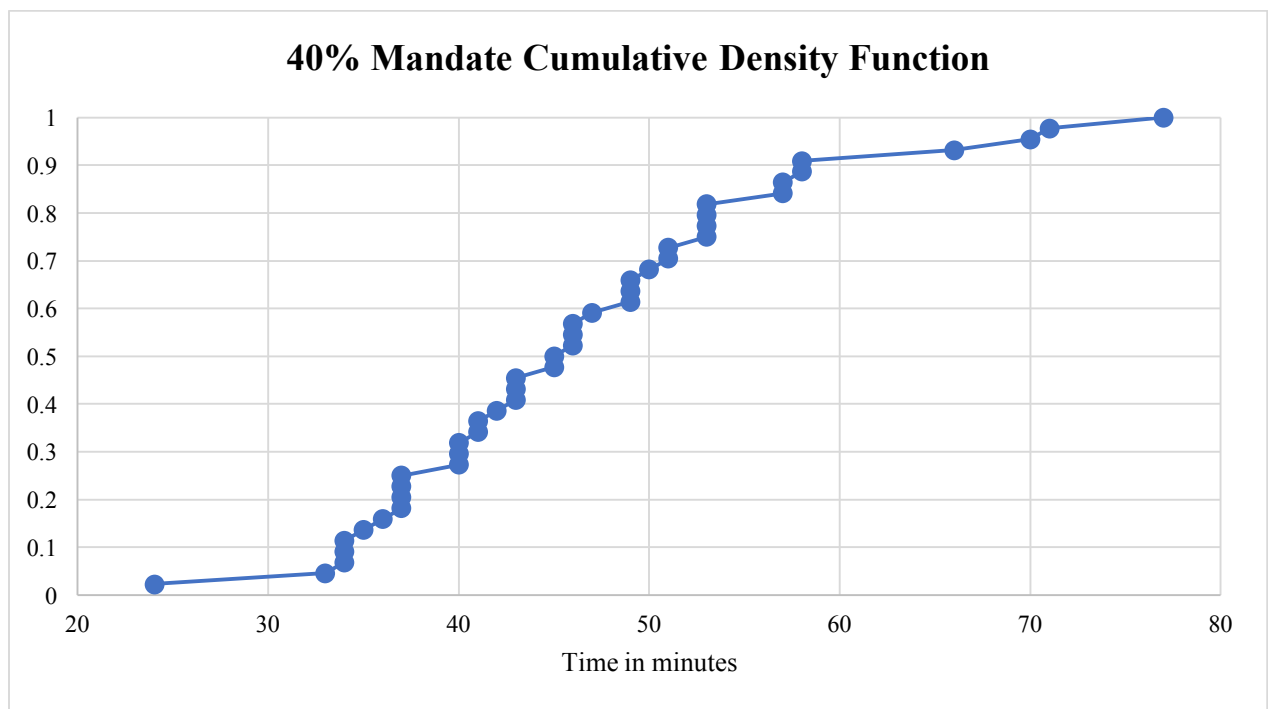


Figure 8: 40% Mandate Cumulative Density Function

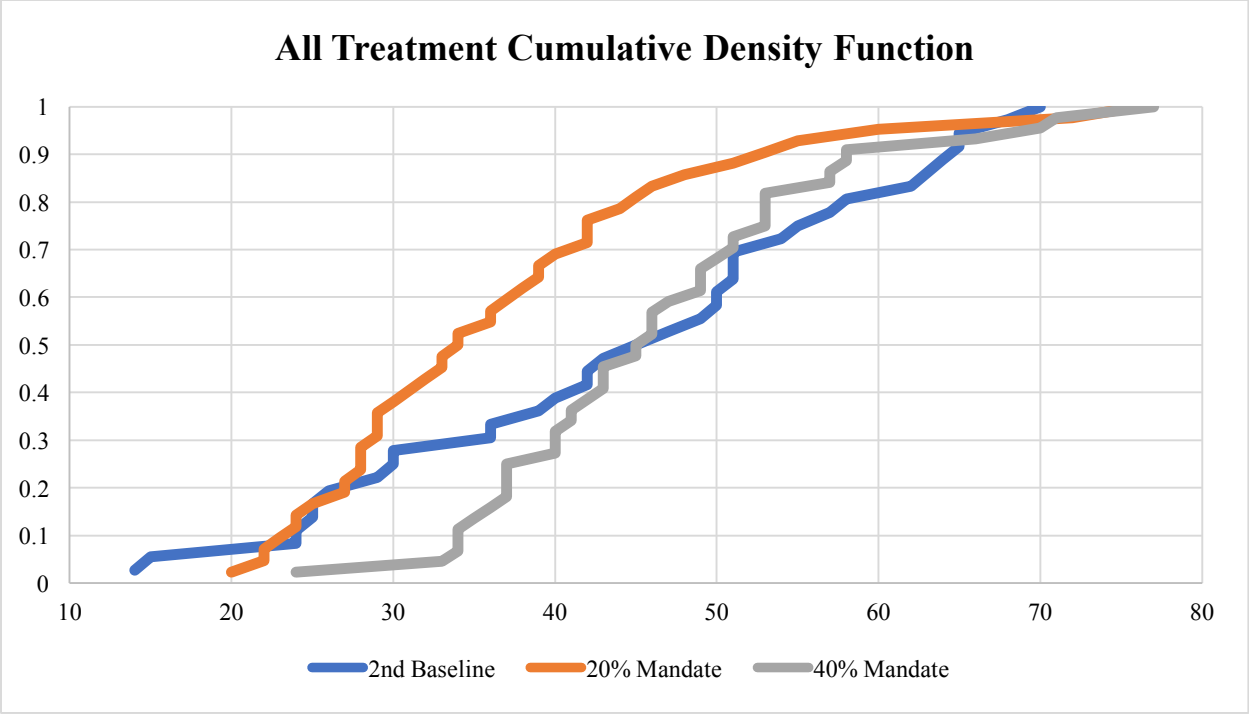


Figure 9: All Treatment Cumulative Density Function

Table 4: Demographic Variables

Independent Variables/Regressions	(1) Total Minutes	(2) Total Minutes	(3) Total Minutes	(4) Total Minutes	(5) Total Minutes	(6) Total Minutes	(7) Total Minutes	(8) Total Minutes	(9) Total Minutes	(10) Total Minutes
Twenty	-7.329** (2.967)	-7.167** (2.973)	-7.500** (2.958)	-7.425** (2.988)	-7.385** (2.975)	-7.378** (2.983)	-7.318** (2.981)	-7.343** (2.994)	-7.275** (3.034)	-7.861** (3.084)
Forty	2.308 (2.936)	2.468 (2.942)	1.841 (2.944)	2.274 (2.948)	2.085 (2.962)	2.361 (2.952)	2.301 (2.948)	2.287 (2.984)	2.383 (3.050)	0.999 (3.156)
Age		0.620 (0.650)								2.007* (1.109)
Female			3.305 (2.381)							3.937 (2.477)
Past Volunteer				1.046 (2.708)						1.678 (2.841)
Liberal					2.006 (3.054)					1.701 (3.233)
White						0.717 (2.394)				0.239 (2.587)
Part Time							-0.724 (5.112)			-1.120 (5.327)
Class Year								-0.0431 (0.921)		-1.959 (1.522)
Religious									-0.243 (2.519)	0.184 (2.633)
Constant	44.31*** (2.177)	31.81*** (13.29)	42.74*** (2.443)	43.58*** (2.883)	44.03*** (2.223)	43.93*** (2.525)	44.35*** (2.205)	44.43*** (3.424)	44.41*** (2.425)	6.040 (19.97)
Observations	122	122	122	122	122	122	122	122	122	122
R-squared	0.095	0.102	0.110	0.097	0.099	0.096	0.096	0.095	0.095	0.141

Ordinary Least Squares Regression

Standard errors in parentheses

*** p<0.01, ** p<0.05, * p<0.1

APPENDIX B

SURVEY MATERIAL

Entry Survey

«Username»

Entry Survey:

Subject ID: econsubject039

Please answer the following questions to the best of your ability.

1. **Sex:**
☐ Male
☐ Female
2. **Age:** _____
3. **Race/ Ethnicity:**
☐ American Indian or other Native American
☐ Asian, Asian American, or Pacific Islander
☐ Black or African American
☐ White (non-Hispanic)
☐ Mexican or Mexican American
☐ Puerto Rican
☐ Other Hispanic or Latino
☐ Multiracial
☐ Other
4. **Student Classification:**
☐ 1st Year
☐ 2nd Year
☐ 3rd Year
☐ 4th Year
☐ 5th Year or greater Undergraduate
☐ Graduate Student
5. **Are you a part time or full time student?**
☐ Part time
☐ Full time
☐ Neither

Entry Survey continued

«Username»

6. How often do you attend religious services?

- ☐ More than once a week
- ☐ Once a week
- ☐ Once a month
- ☐ Only for religious holidays
- ☐ Never

7. What are your weekly leisure spending habits (dollar amount)?

- ☐ \$0 – 20
- ☐ \$20– 40
- ☐ \$40 – 60
- ☐ \$60 – 80
- ☐ \$80 – 100
- ☐ \$100 - 120
- ☐ \$120 – 140
- ☐ More than \$140

Figure 10: Entry Survey

Exit Survey

Exit Survey

Subject ID: _____

TO: _____

1. Prior to this experiment, have you volunteered for charity in the past?

☐ Yes

☐ No

If you answered yes to the question above, please explain.

2. Have you ever donated money to a charitable organization?

☐ Yes

☐ No

If yes, approximately how much did you donate in the last 12 months?

3. Are you associated with an organization that requires you to volunteer?

☐ Yes

☐ No

If you answered yes to the question above, please explain.

4. Have you participated in Texas A&M's "The Big Event" in the past?

☐ Yes

☐ No

Exit Survey continued

5. Will you be attending The Big Event this school year?

☐ Yes
☐ No
☐ Undecided

Why or why not?

6. Upon leaving this experiment, what is your next time commitment? (Class or other appointment)

7. How many economic courses have you taken?

☐ 0
☐ 1 - 2
☐ 3 - 4
☐ 5 - 6
☐ 7 or more

8. Which of the following best describes your political ideology?

☐ Left/Liberal
☐ Right/Conservative
☐ Centrist/Moderate
☐ Abstain/Apolitical/Other _____

9. How would you rate your experience with Freerice?

☐ Very positive
☐ Positive
☐ Unremarkable
☐ Negative
☐ Very negative

Exit Survey continued

10. What do you think about Freerice's charity model?

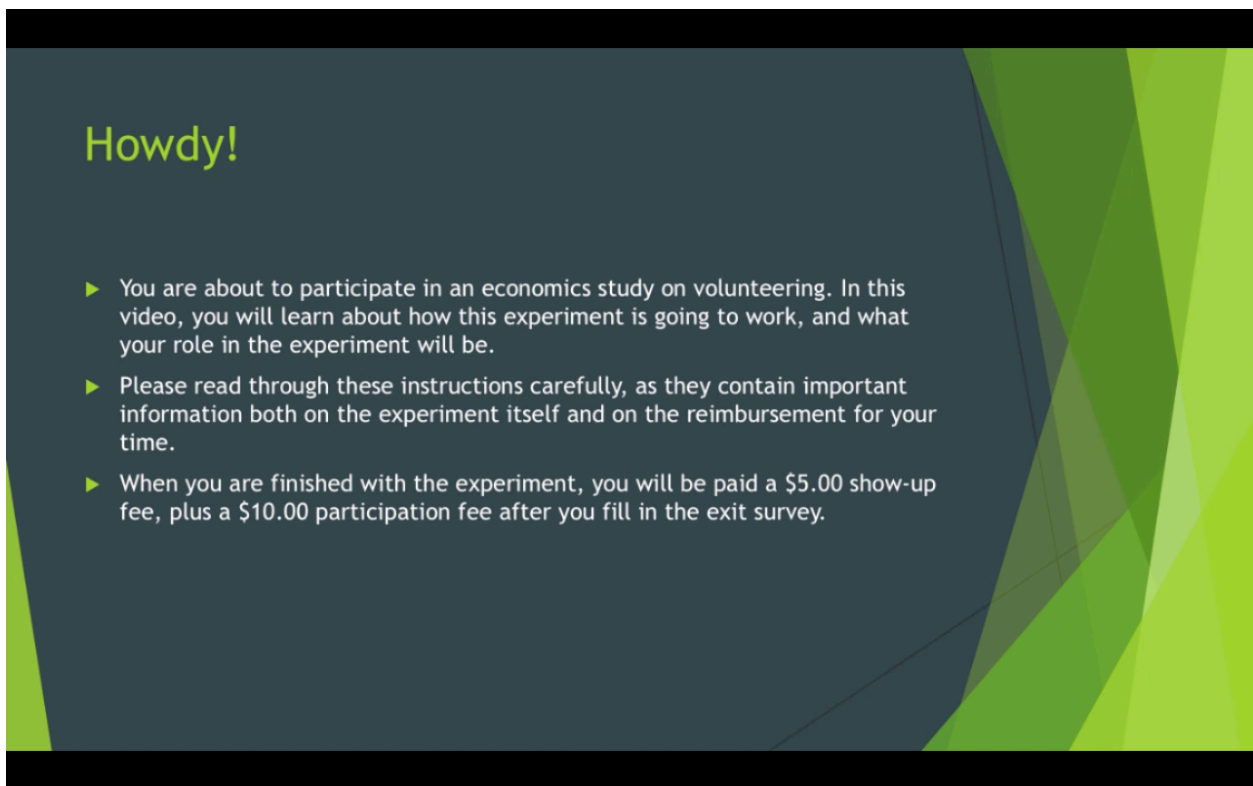
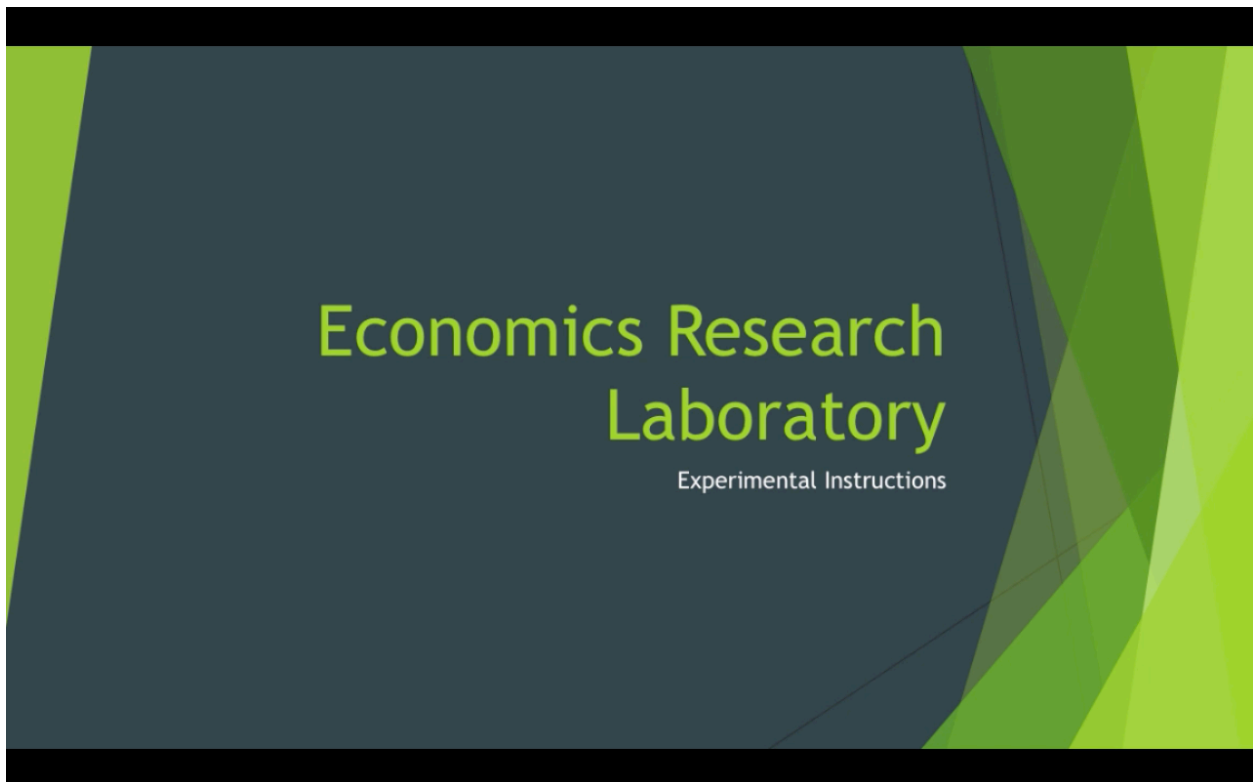
- ☐ Very good
- ☐ Good
- ☐ Unremarkable
- ☐ Bad
- ☐ Very bad

11. Did you come in with a friend? If so, how many?

- ☐ I did not come in with a friend.
- ☐ 1
- ☐ 2
- ☐ 3+

Figure 11: Exit Survey

Original Baseline Video



Original Baseline Video continued

Charity

- ▶ In this experiment, you will be volunteering for a charity, Free Rice.
- ▶ Free Rice is an initiative of the United Nations World Food Programme, and has as its goal both providing education and addressing the needs of the hungry across the world.
- ▶ Your job volunteering for Free Rice consists of answering a series of educational questions.
- ▶ For every question you answer correctly, ten grains of rice are donated by the UN World Food Programme to those in need.
- ▶ The rice is paid for by Freerice's partner organizations, whose names may appear throughout the website.



Logging In

- ▶ You have been assigned a random, anonymous username and password which will only be used for this experiment. One of our experimenters has already signed in for you using these credentials.
- ▶ If you have played Free Rice before, DO NOT log in under your personal account.
- ▶ The gameplay page has been pre-loaded on your computer, but if you have trouble starting the game, or if you are accidentally logged out, raise your hand, and an experimenter will come to assist you.

Original Baseline Video continued

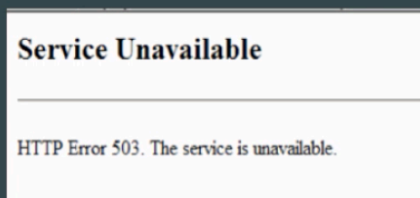
Playing the Game

- ▶ When you are logged in, you will see a screen similar to the one on your right.
- ▶ The game is played in the interface box here.
- ▶ Answer the question correctly, and ten grains of rice will be donated to the United Nations World Food Programme.
- ▶ If you wish, you can also switch question subjects. Just click the button here, and choose your new topic.



Errors

- ▶ Occasionally, the internet may slow down in the lab, or we may experience a bad connection.
- ▶ Sometimes, this will slow down the game response time, and more rarely, it might present you with a "503 Error" message, like this.
- ▶ If this occurs, do not worry. Just press the refresh button at the top of your browser, and the error should resolve itself shortly.
- ▶ Refreshing the page DOES NOT affect your rice totals.
- ▶ If you continue to have problems, raise your hand, and an experimenter will come to assist you.



Original Baseline Video continued

Lab Regulations

- ▶ While you are in the lab, you are allowed to listen to music on your cell phone, iPod, or MP3 player using your earphones, but we request that you not talk to your fellow volunteers.
- ▶ When you are ready to leave, raise your hand, and an experimenter will come to your desk, and bring you an exit survey. He or she will then ask you to fill out this survey and bring it, along with the entry survey you took before this video, to the experimenter stationed outside the lab at the payment window.
- ▶ Do NOT begin playing the game without first logging in. This step is critical for data collection purposes. If you are accidentally logged out, raise your hand, and an experimenter will come to assist you.

Lab Regulations

- ▶ After filling out the exit survey, you will be paid for your participation in the experiment.
- ▶ Remember, your payment does NOT depend on either the time you volunteered or the quantity of rice you have donated. ALL volunteers will be paid the same amount, \$15.00 (\$10 plus a \$5 show-up fee).
- ▶ In order to receive the \$10 however, you are required to fill out the exit survey before you leave. This is the only factor that plays into your final payment.

Original Baseline Video continued

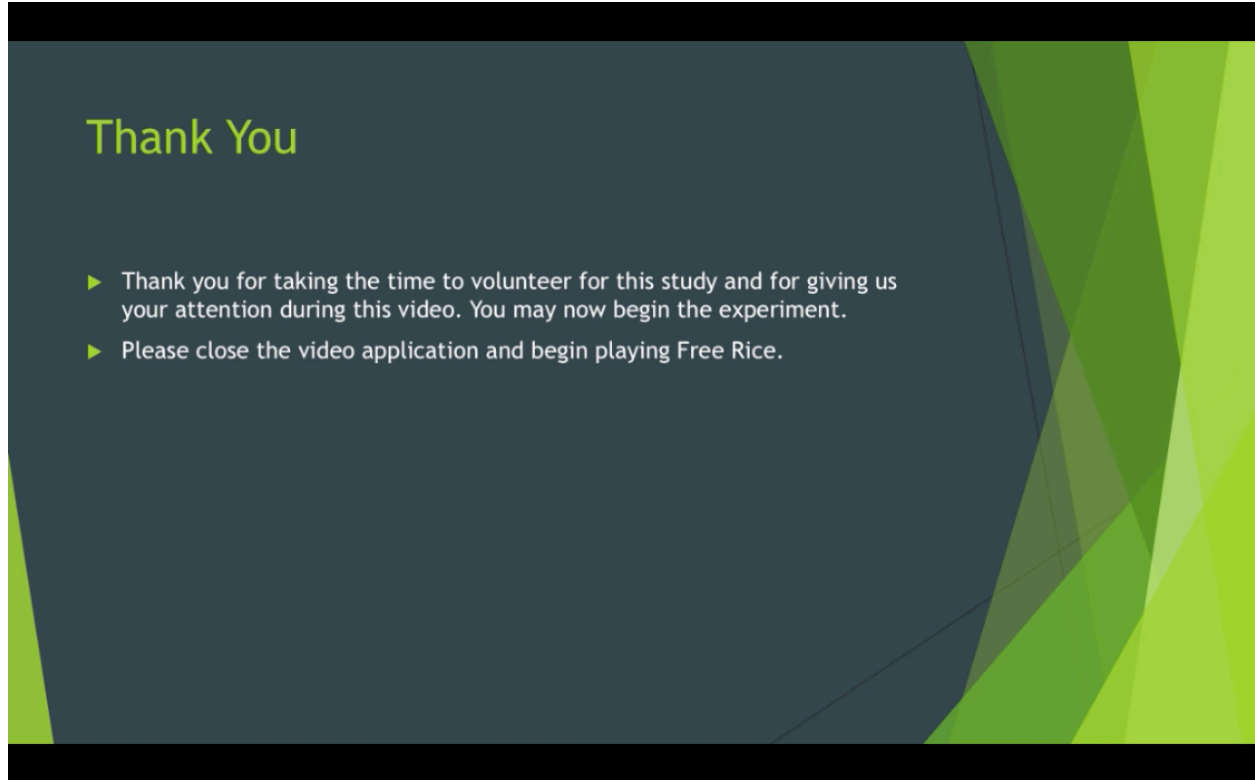
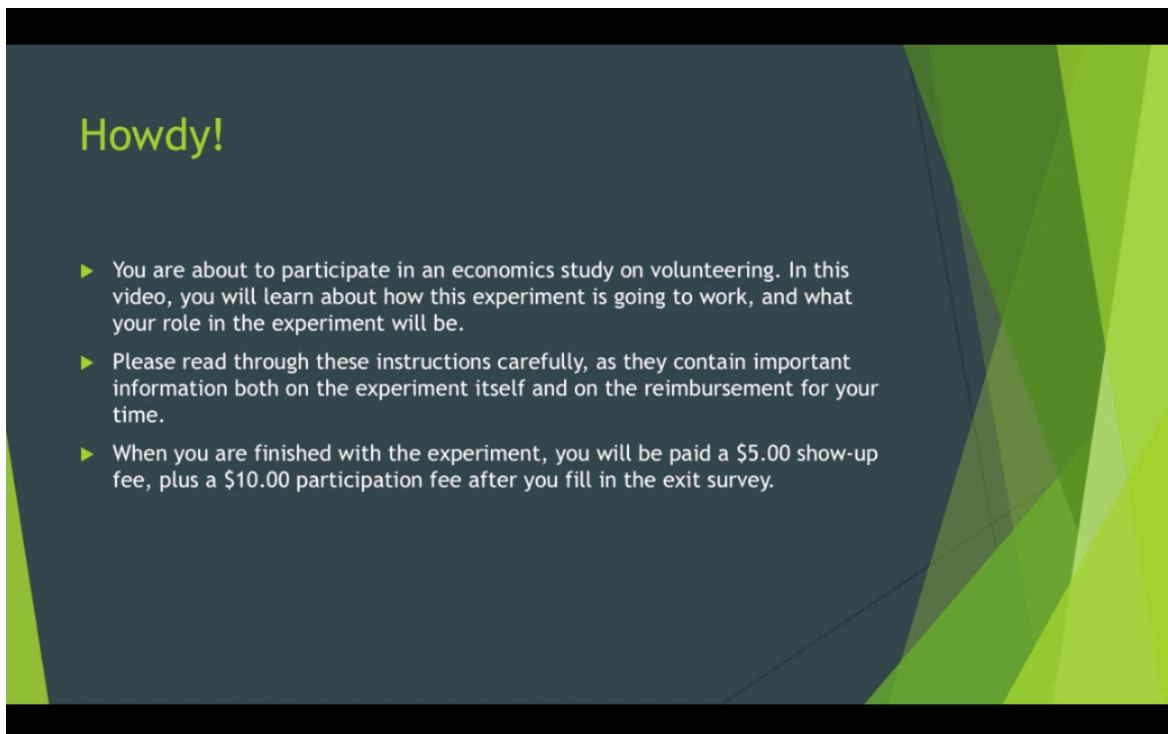
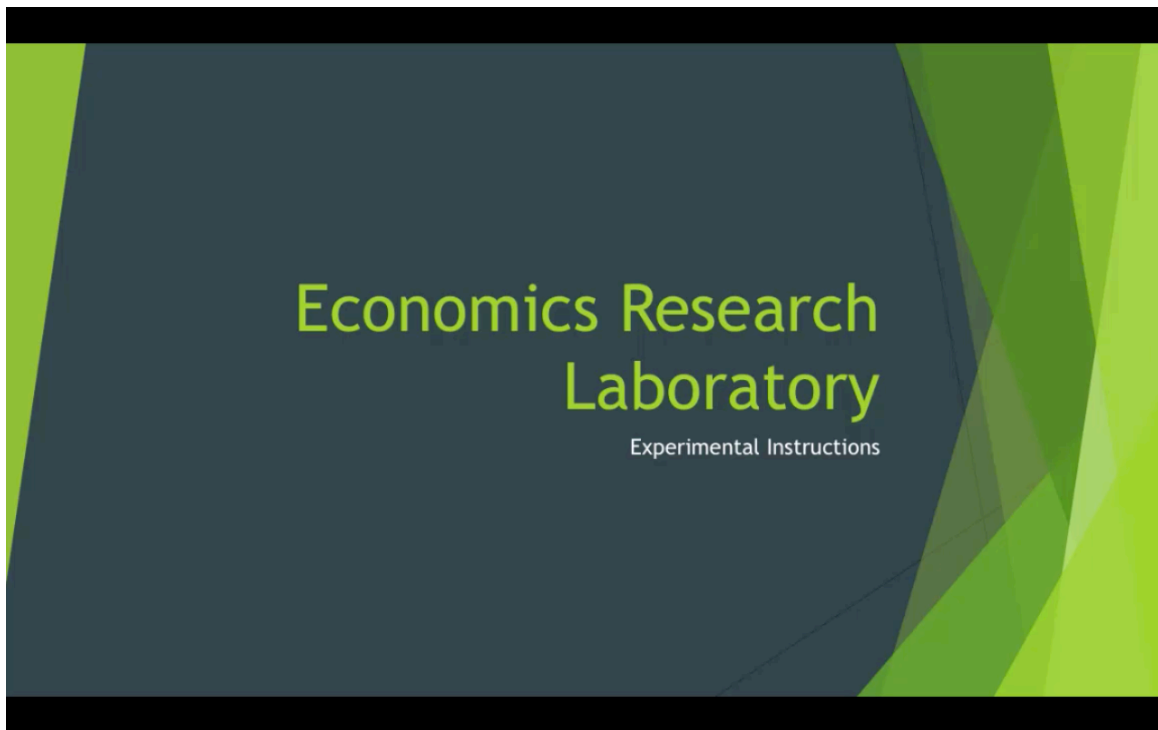


Figure 12: Original Baseline Video

New Baseline Video



New Baseline Video continued

Charity

- ▶ In this experiment, you will be volunteering for a charity, Free Rice.
- ▶ Free Rice is an initiative of the United Nations World Food Programme, and has as its goal both providing education and addressing the needs of the hungry across the world.
- ▶ Your job volunteering for Free Rice consists of answering a series of educational questions.
- ▶ For every question you answer correctly, ten grains of rice are donated by the UN World Food Programme to those in need.
- ▶ The rice is paid for by Freerice's partner organizations, whose names may appear throughout the website.

The logo for Free Rice, featuring the words "FREE" and "Rice" in a white, sans-serif font. "FREE" is in all caps and smaller, while "Rice" is in title case and larger. They are positioned on a bright green rectangular background.

Logging In

- ▶ You have been assigned a random, anonymous username and password which will only be used for this experiment. One of our experimenters has already signed in for you using these credentials.
- ▶ If you have played Free Rice before, DO NOT log in under your personal account.
- ▶ The gameplay page has been pre-loaded on your computer, but if you have trouble starting the game, or if you are accidentally logged out, raise your hand, and an experimenter will come to assist you.

New Baseline Video continued

Playing the Game

- ▶ When you are logged in, you will see a screen similar to the one on your right.
- ▶ The game is played in the interface box here.
- ▶ Answer the question correctly, and ten grains of rice will be donated to the United Nations World Food Programme.
- ▶ If you wish, you can also switch question subjects. Just click the button here, and choose your new topic.



Errors

- ▶ Occasionally, the internet may slow down in the lab, or we may experience a bad connection.
- ▶ Sometimes, this will slow down the game response time, and more rarely, it might present you with a "503 Error" message, like this.
- ▶ If this occurs, do not worry. Just press the refresh button at the top of your browser, and the error should resolve itself shortly.
- ▶ Refreshing the page DOES NOT affect your rice totals.
- ▶ If you continue to have problems, raise your hand, and an experimenter will come to assist you.



New Baseline Video continued

Lab Regulations

- ▶ Do NOT begin playing the game without first logging in. This step is critical for data collection purposes. If you are accidentally logged out, raise your hand, and an experimenter will come to assist you.
- ▶ While you are in the lab, you are allowed to listen to music on your cell phone, iPod, or MP3 player using your earphones, but we request that you not talk to your fellow volunteers.
- ▶ Freerice is an external website, so when you are ready to leave, just stop answering questions and raise your hand.

Lab Regulations

- ▶ After filling out the exit survey, you will be paid for your participation in the experiment.
- ▶ Remember, your payment does NOT depend on either the time you volunteered or the quantity of rice you have donated. ALL volunteers will be paid the same amount, \$15.00 (\$10 plus a \$5 show-up fee).
- ▶ In order to receive the \$10 however, you are required to fill out the exit survey before you leave. This is the only factor that plays into your final payment.

New Baseline Video continued

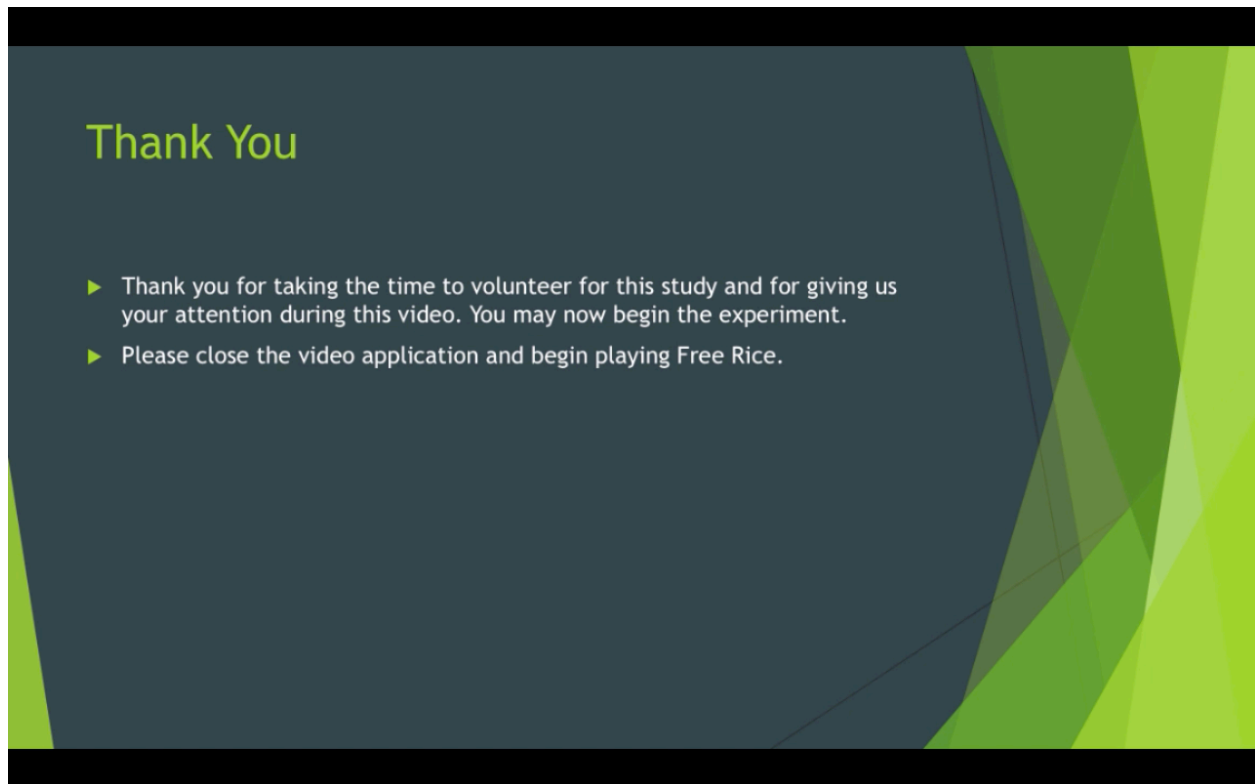
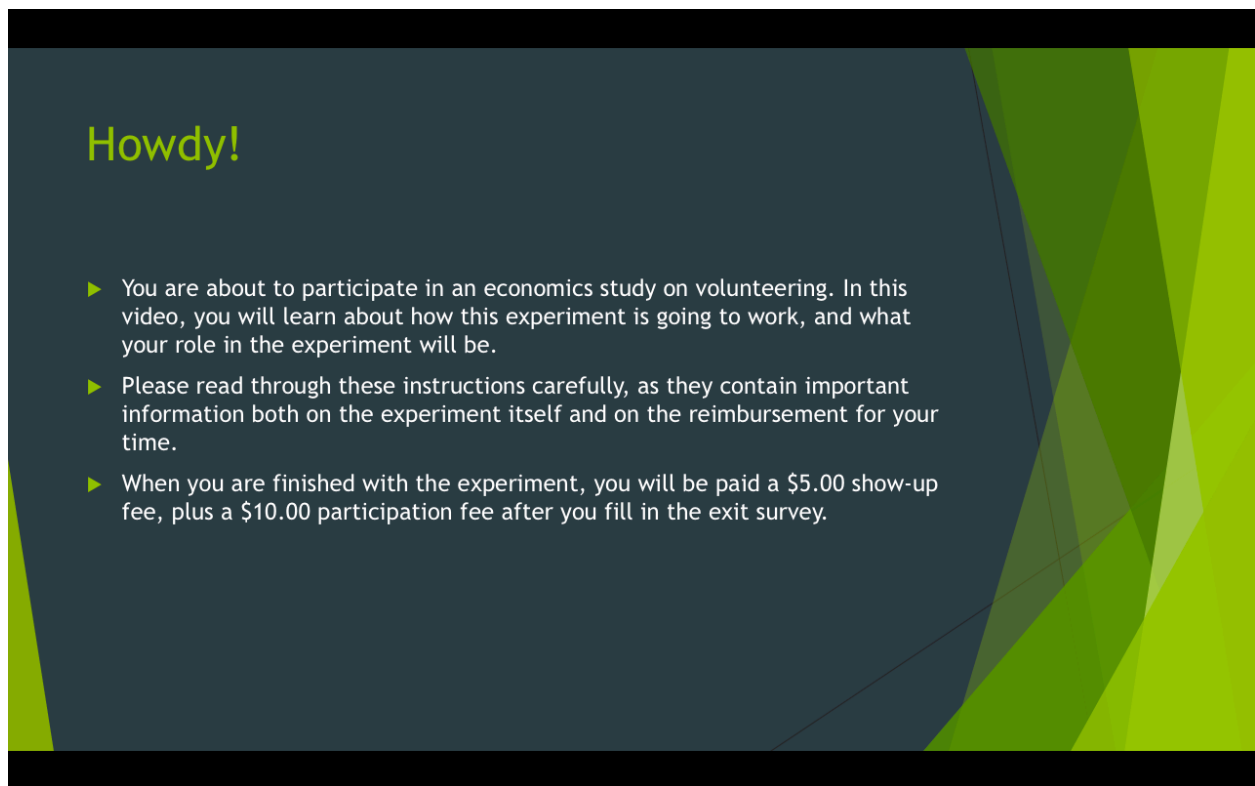
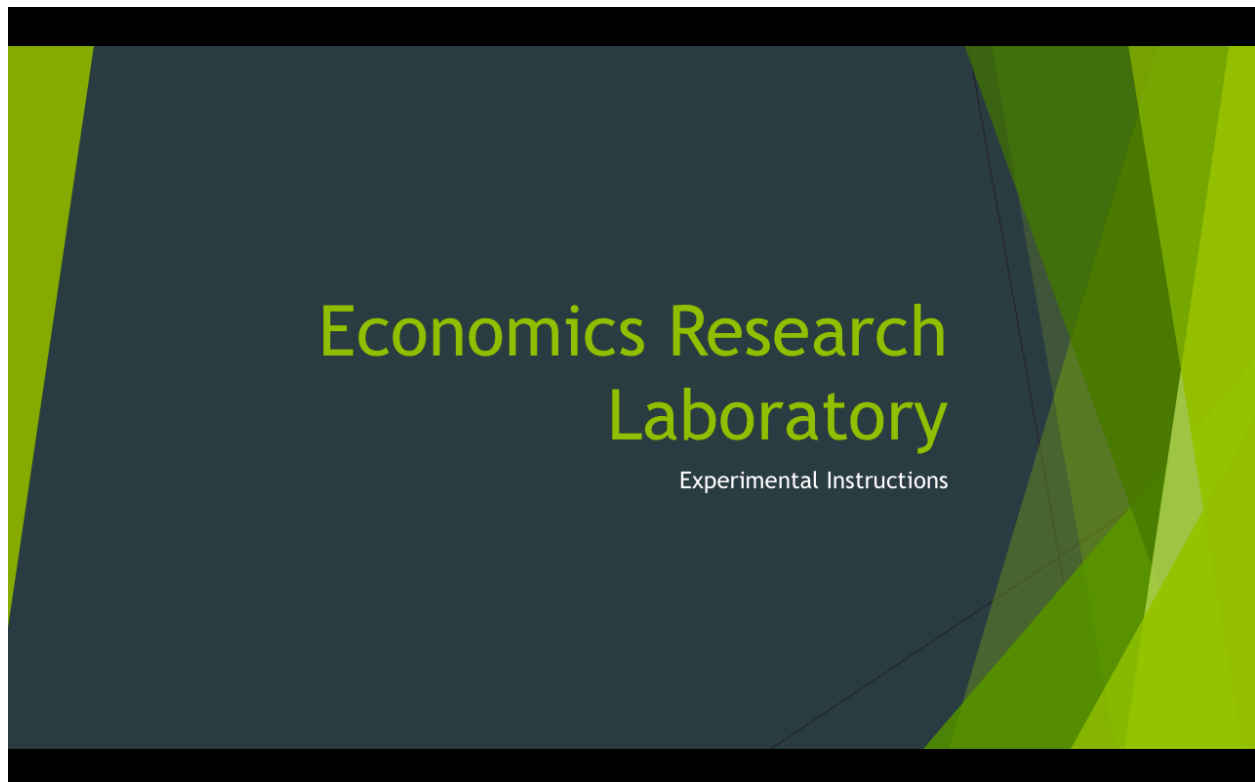


Figure 13: New Baseline Video

20% Mandate Video



20% Mandate Video continued

Charity

- ▶ In this experiment, you will be volunteering for a charity, Free Rice.
- ▶ Free Rice is an initiative of the United Nations World Food Programme, and has as its goal both providing education and addressing the needs of the hungry across the world.
- ▶ Your job volunteering for Free Rice consists of answering a series of educational questions.
- ▶ For every question you answer correctly, ten grains of rice are donated by the UN World Food Programme to those in need.
- ▶ The rice is paid for by Freerice's partner organizations, whose names may appear throughout the website.



Logging In

- ▶ You have been assigned a random, anonymous username and password which will only be used for this experiment. One of our experimenters has already signed in for you using these credentials.
- ▶ If you have played Free Rice before, DO NOT log in under your personal account.
- ▶ The gameplay page has been pre-loaded on your computer, but if you have trouble starting the game, or if you are accidentally logged out, raise your hand, and an experimenter will come to assist you.

20% Mandate Video continued

Playing the Game

- ▶ When you are logged in, you will see a screen similar to the one on your right.
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- ▶ If you wish, you can also switch question subjects. Just click the button here, and choose your new topic.



Errors

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- ▶ Refreshing the page DOES NOT affect your rice totals.
- ▶ If you continue to have problems, raise your hand, and an experimenter will come to assist you.



20% Mandate Video continued

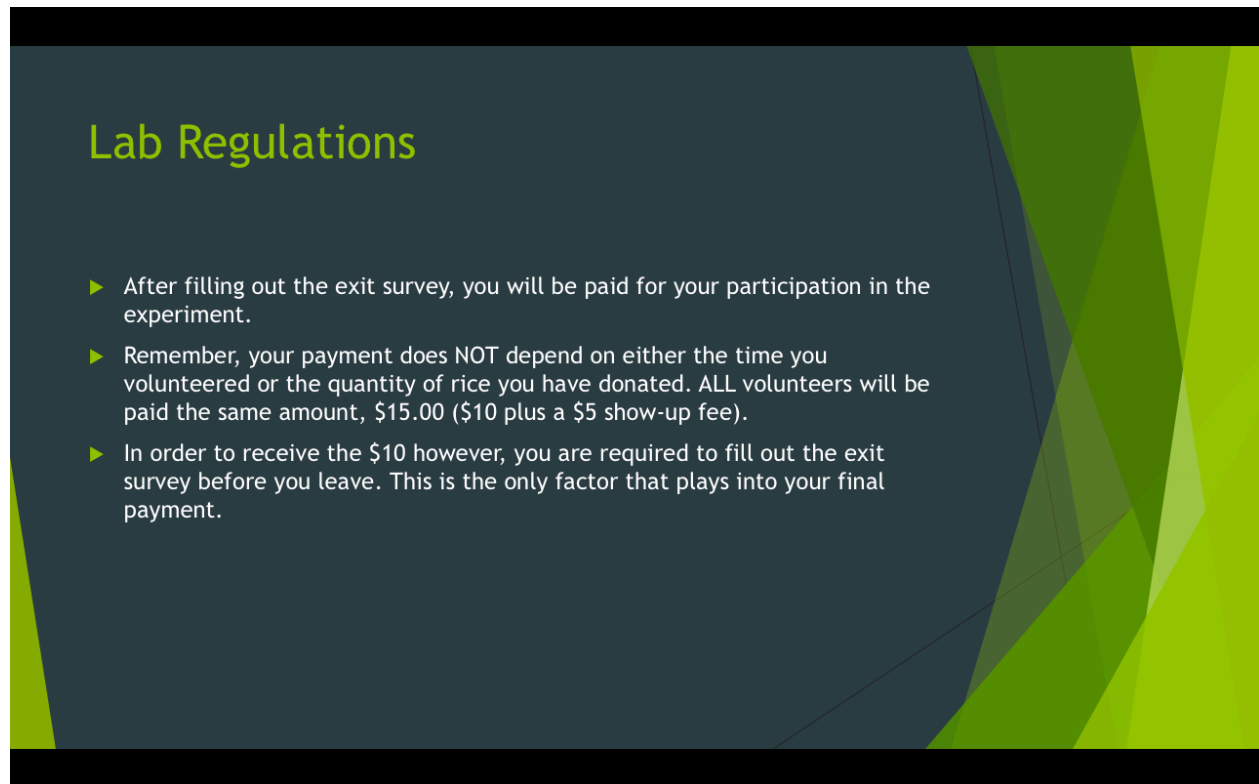
Volunteering Request

- ▶ Although you may volunteer for longer if you like, in this session, we ask that you volunteer for at least 12 minutes before you leave.
- ▶ When you are ready to leave, raise your hand, and an experimenter will come to your desk, and bring you an exit survey. He or she will then ask you to fill out this survey and bring it, along with the entry survey you took before this video, to the experimenter stationed outside the lab at the payment window.

Lab Regulations

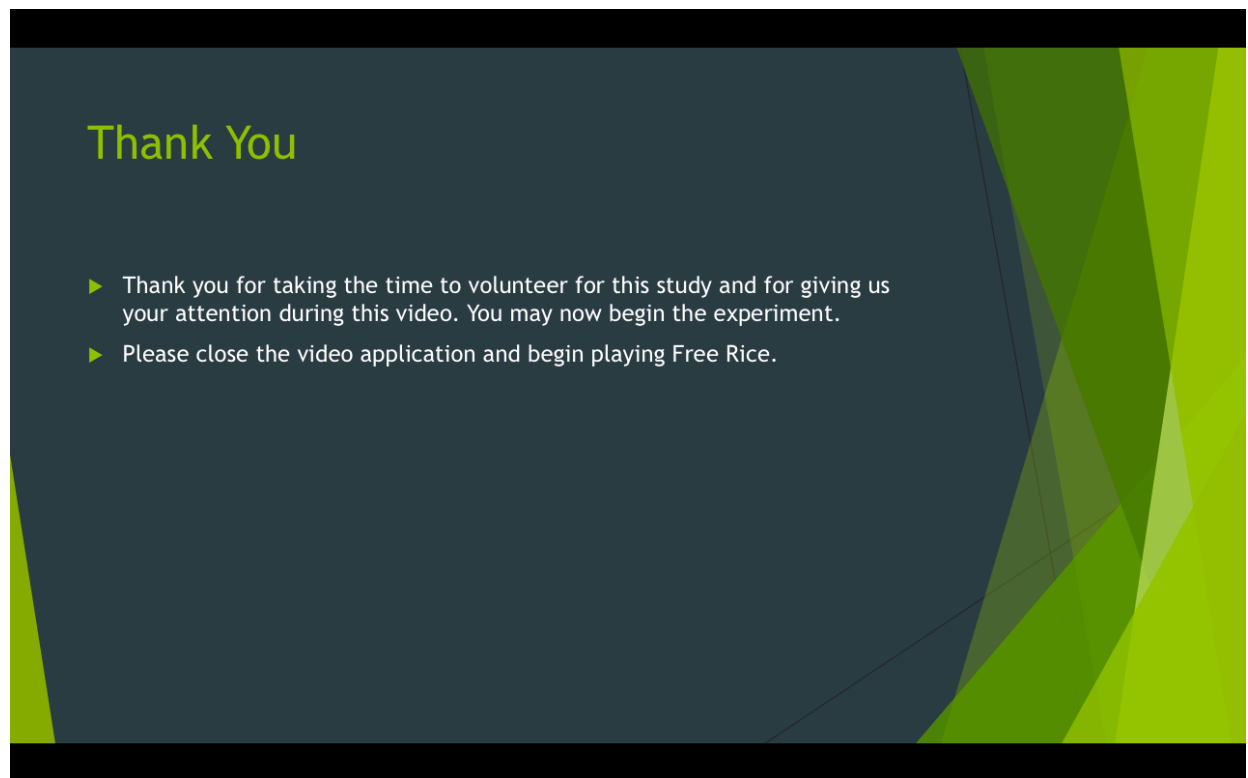
- ▶ While you are in the lab, you are allowed to listen to music on your cell phone, iPod, or MP3 player using your earphones, but we request that you not talk to your fellow volunteers.
- ▶ Do NOT begin playing the game without first logging in. This step is critical for data collection purposes. If you are accidentally logged out, raise your hand, and an experimenter will come to assist you.

20% Mandate Video continued

A presentation slide with a dark blue background and a green geometric pattern on the right side. The title 'Lab Regulations' is in green. The content is a bulleted list of three items in white text.

Lab Regulations

- ▶ After filling out the exit survey, you will be paid for your participation in the experiment.
- ▶ Remember, your payment does NOT depend on either the time you volunteered or the quantity of rice you have donated. ALL volunteers will be paid the same amount, \$15.00 (\$10 plus a \$5 show-up fee).
- ▶ In order to receive the \$10 however, you are required to fill out the exit survey before you leave. This is the only factor that plays into your final payment.

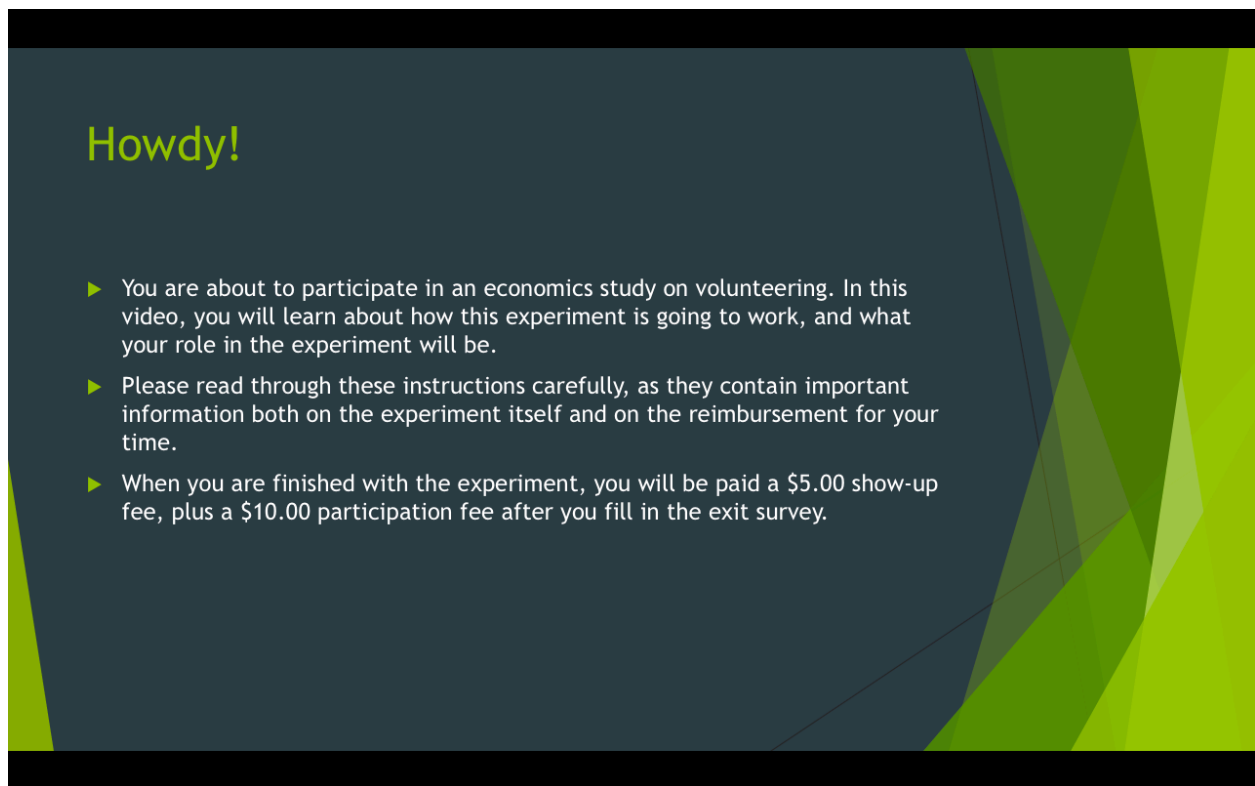
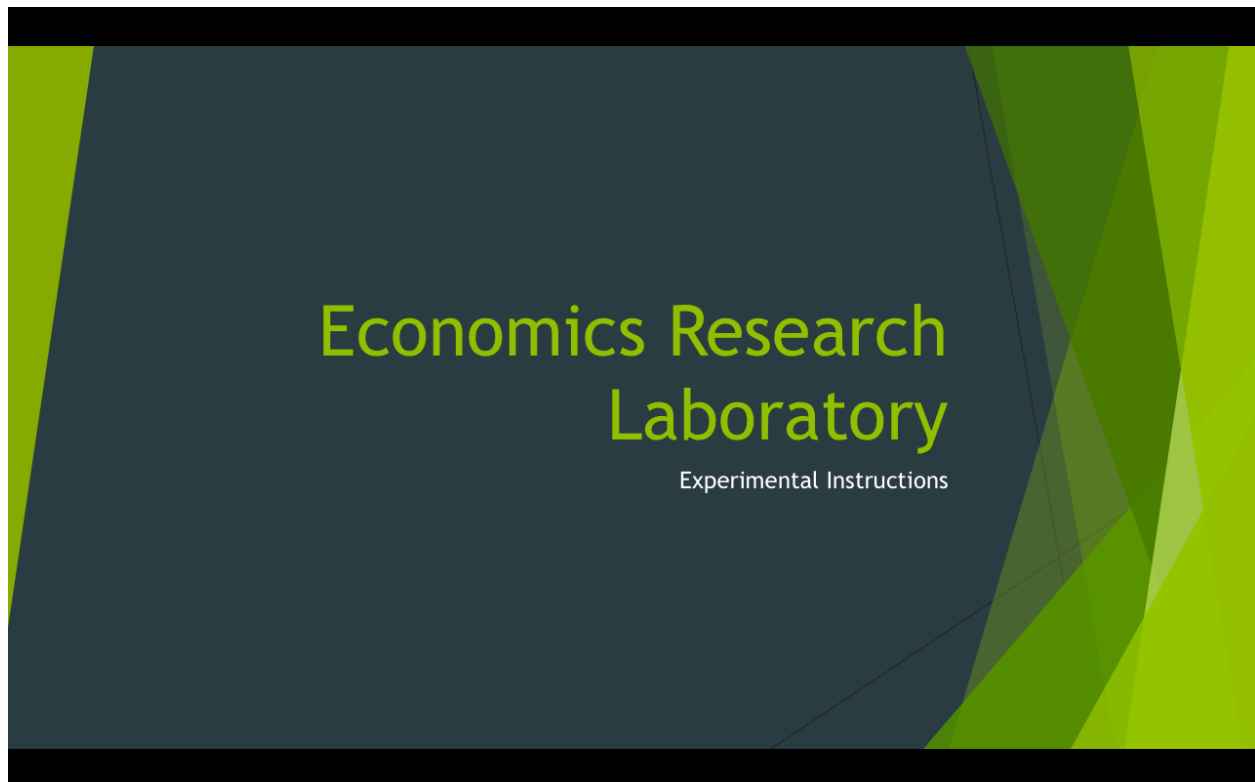
A presentation slide with a dark blue background and a green geometric pattern on the right side. The title 'Thank You' is in green. The content is a bulleted list of two items in white text.

Thank You

- ▶ Thank you for taking the time to volunteer for this study and for giving us your attention during this video. You may now begin the experiment.
- ▶ Please close the video application and begin playing Free Rice.

Figure 14: 20% Baseline Video

40% Mandate Video



40% Mandate Video continued

Charity

- ▶ In this experiment, you will be volunteering for a charity, Free Rice.
- ▶ Free Rice is an initiative of the United Nations World Food Programme, and has as its goal both providing education and addressing the needs of the hungry across the world.
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40% Mandate Video continued

Playing the Game

- ▶ When you are logged in, you will see a screen similar to the one on your right.
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- ▶ If this occurs, do not worry. Just press the refresh button at the top of your browser, and the error should resolve itself shortly.
- ▶ Refreshing the page DOES NOT affect your rice totals.
- ▶ If you continue to have problems, raise your hand, and an experimenter will come to assist you.



40% Mandate Video continued

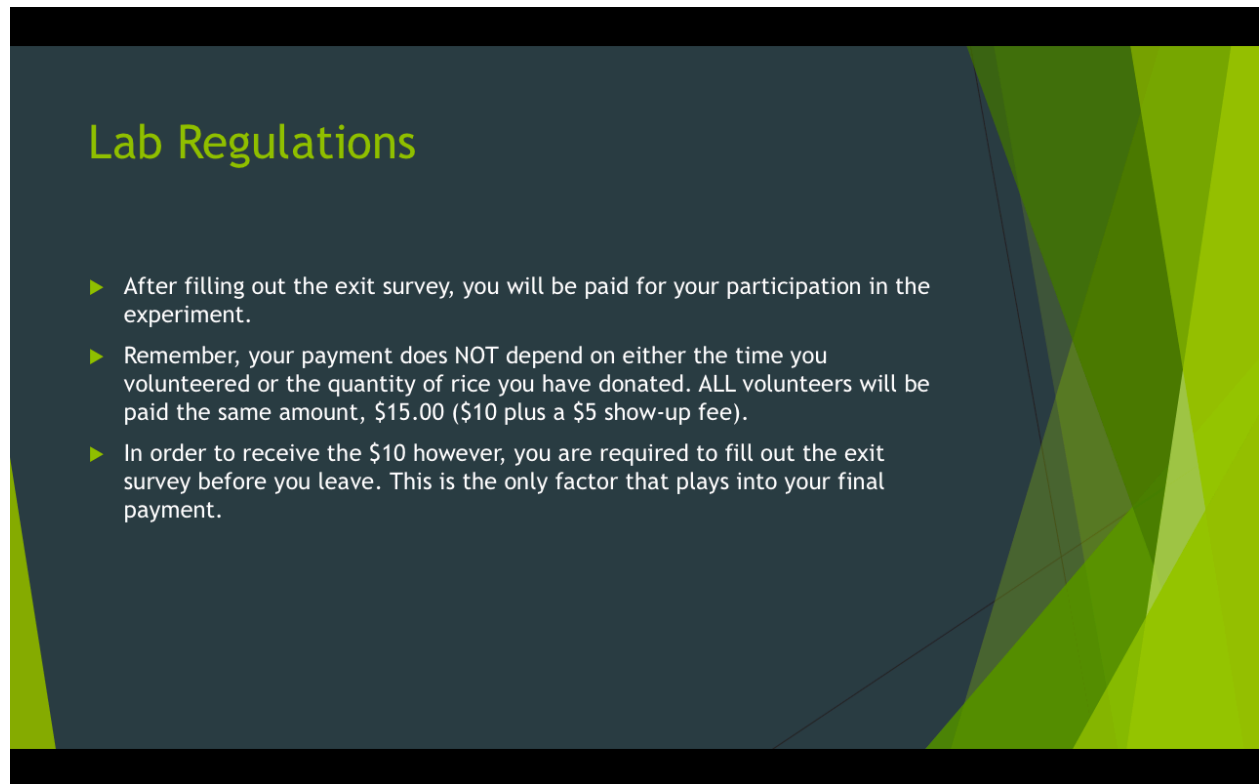
Volunteering Request

- ▶ Although you may volunteer for longer if you like, in this session, we ask that you volunteer for at least 24 minutes before you leave.
- ▶ When you are ready to leave, raise your hand, and an experimenter will come to your desk, and bring you an exit survey. He or she will then ask you to fill out this survey and bring it, along with the entry survey you took before this video, to the experimenter stationed outside the lab at the payment window.

Lab Regulations

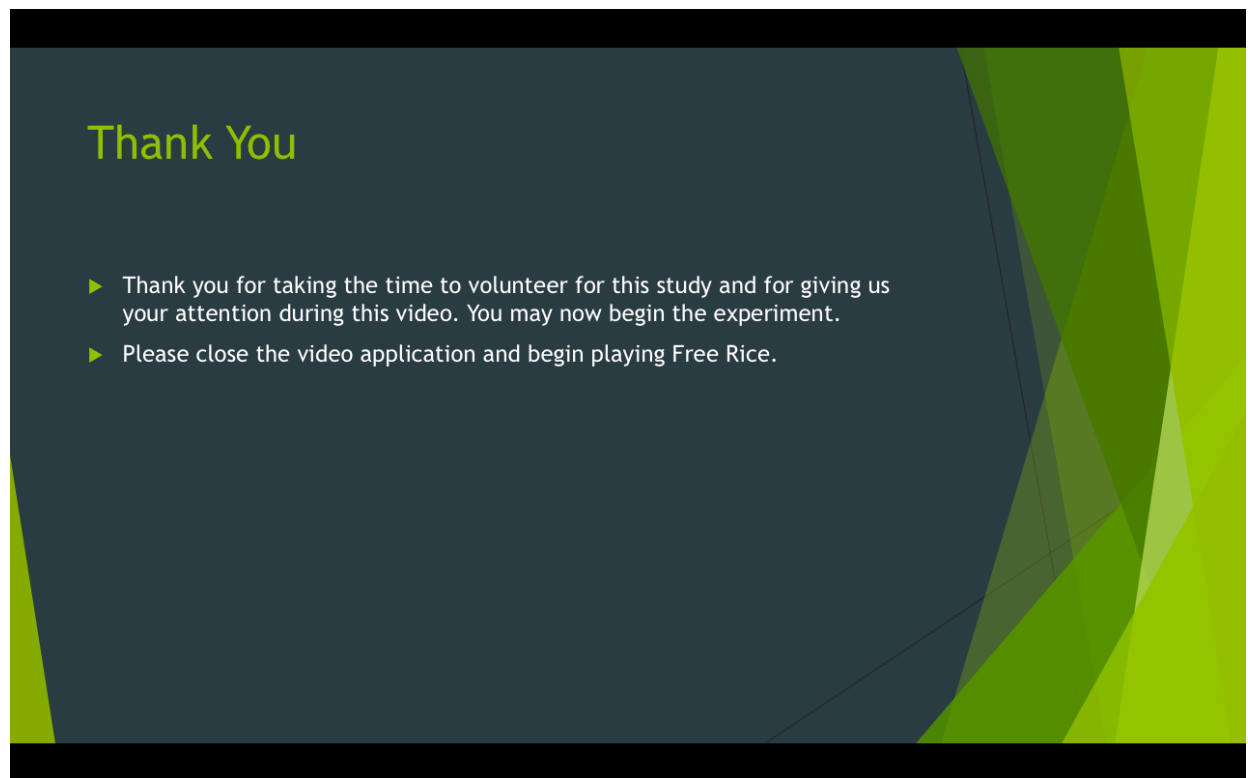
- ▶ While you are in the lab, you are allowed to listen to music on your cell phone, iPod, or MP3 player using your earphones, but we request that you not talk to your fellow volunteers.
- ▶ Do NOT begin playing the game without first logging in. This step is critical for data collection purposes. If you are accidentally logged out, raise your hand, and an experimenter will come to assist you.

40% Mandate Video continued

A presentation slide with a dark blue background and a green geometric pattern on the right side. The title 'Lab Regulations' is in green. The content is a bulleted list of three items in white text.

Lab Regulations

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- ▶ Remember, your payment does NOT depend on either the time you volunteered or the quantity of rice you have donated. ALL volunteers will be paid the same amount, \$15.00 (\$10 plus a \$5 show-up fee).
- ▶ In order to receive the \$10 however, you are required to fill out the exit survey before you leave. This is the only factor that plays into your final payment.

A presentation slide with a dark blue background and a green geometric pattern on the right side. The title 'Thank You' is in green. The content is a bulleted list of two items in white text.

Thank You

- ▶ Thank you for taking the time to volunteer for this study and for giving us your attention during this video. You may now begin the experiment.
- ▶ Please close the video application and begin playing Free Rice.

Figure 15: 40% Mandate

